IS883: Deploying Generative AI

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Financial Modeling of Gen AI Applications

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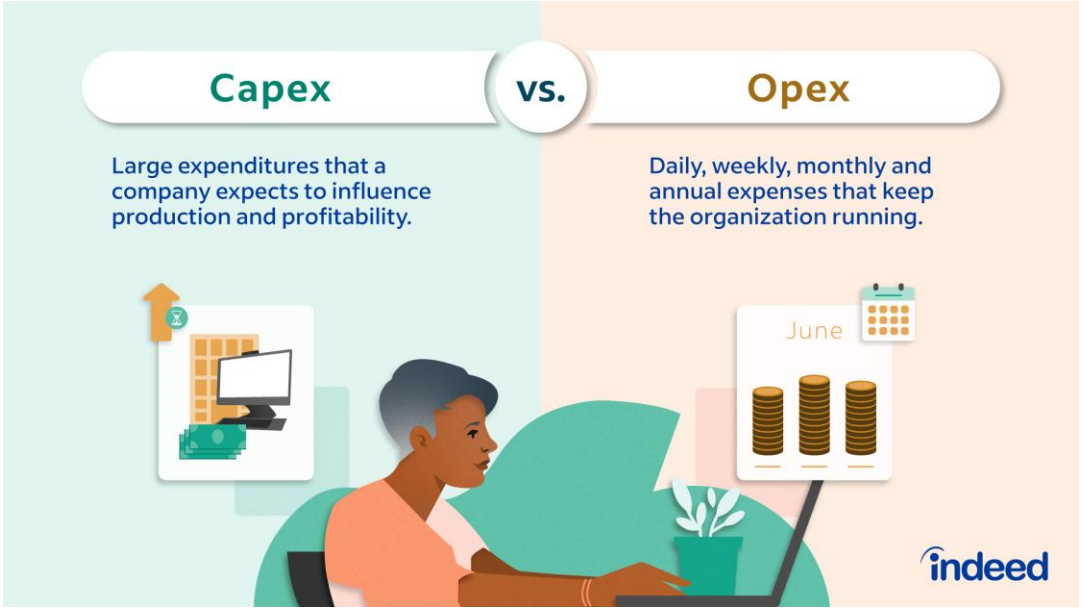
Cost-Benefit Analysis

- When building a Gen AI solution or application, we need to assess its ***viability***.
- We generally would look at the ***ROI*** (Return on Investment)

$$\text{ROI} = \frac{\text{Net Profit}}{\text{Total Cost}} = \frac{\text{Total Revenue} - \text{Total Cost}}{\text{Total Cost}}$$

Total Cost

- Total cost can be broken down into two types:
  - **Upfront Costs (CapEx):** Do not repeat.
  - **Ongoing Costs (OpEx):** Recurring expenses to keep the service running.



Upfront Costs

- Examples



Infrastructure



Data



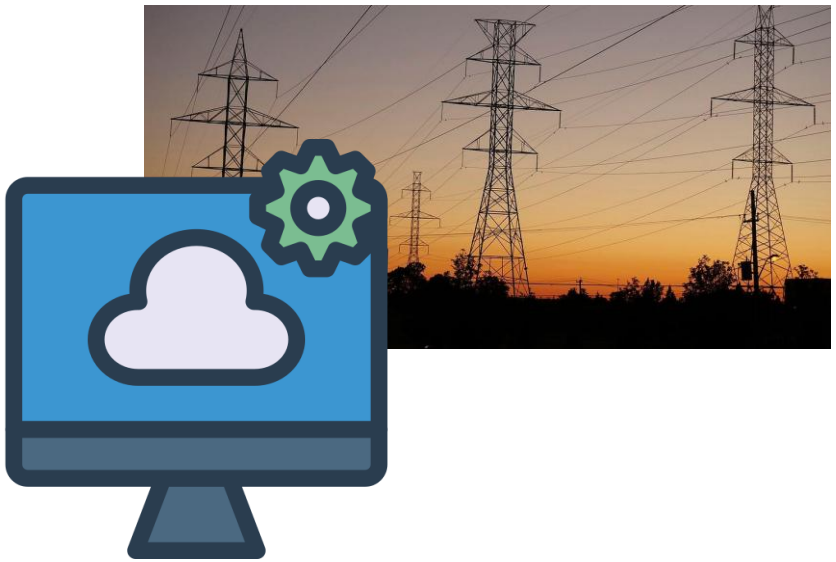
Software and Tools



Staffing

Ongoing Costs

- Examples



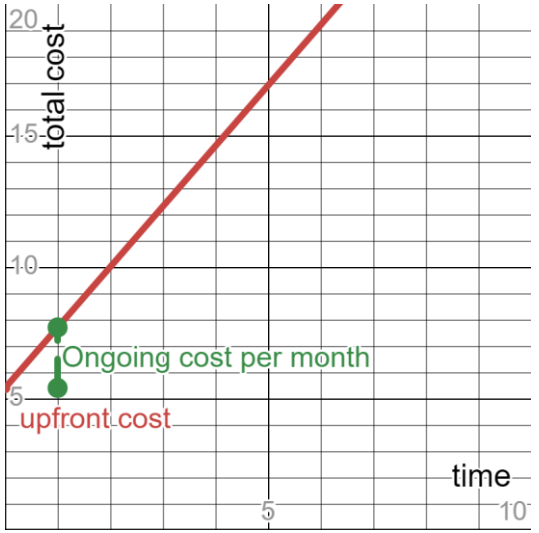
Operational costs (Energy, subscriptions, etc.)



Maintenance costs

Why Care For This Distinction?

- Because that's what matters for the cost equation.
  - Your decision should be based on:
    - Cashflow: **How much** money you have and **when**.
    - Product Lifespan.
    - Scalability and Flexibility: Pay-as-you-go provides more control and easier adaptation.
    - Risk: Lower upfront commitment is more risk averse.



$$\text{Total Cost} = \text{Upfront Costs} + (\text{Ongoing Cost per Unit of Time} \times \text{Time Period})$$

What Affects Cost in Gen AI?

In-house Approach	Pay-as-you-go Approach
• **Infrastructure** (upfront or ongoing). • **Software licensing** (ongoing) • **Developers and Staff** (upfront and ongoing). • **Maintenance, energy, and upgrades** (ongoing). • Includes model size (computational needs) • **Training cost** (upfront or ongoing): • Data size	• **Developers and Staff** (upfront and ongoing). • **Cloud subscription fees** (ongoing). • **Usages** (ongoing): • Number of tokens (i.e., request and response sizes). • Model type.

Total Revenue

- Total revenue depends on the provided service/product, but could be measured as
  - **Cost Savings:** If product or service is to be used internally, how much cost reduction is there due to automation and increased efficiency?
  - **Sales Increase:** due to an increase in customer satisfaction.

Evaluations: ROI, CBR, and Break-Even Analysis

- Break-Even Point: Time it takes for Total Revenue to become equal Total Cost.
- Cost-Benefit Ratio is simply: $\mathrm{CBR} = \mathrm{ROI} + 1$
  - ROI is negative when product/service is not viable.
  - CBR is less than one when product/service is not viable.

- Try it here!

Resources

- Gen AI: too much spend, too little benefit?
- Understanding the Cost of LLMs